Complex analysis toolkit. Evaluating the momentum integrals that arise from the spectral representation requires contour integration. Here we collect the key results.

0. **Analytic functions.** A complex function $f(z)$ is analytic or holomorphic in a region if it is complex-differentiable at every point of that region, i.e., the limit

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} \tag{1}$$

exists and is independent of the direction in which $\Delta z \to 0$ in the complex plane. Analytic functions can be expanded in a convergent power series (Taylor series) around any point in their domain. Points where $f(z)$ fails to be analytic are called singularities, the simplest type of which is a pole such as $f(z) \sim (z - z_0)^{-1}$. As you can guess from the spectral decomposition above, the Green's function has poles at the energy eigenvalues E_n, so $G^{\pm}(E)$ will be analytic everywhere along the real line E except at the locations of the physical energy eigenvalues E_n.

1. **Cauchy's residue theorem.** If $f(z)$ is analytic inside and on a closed contour $\mathcal{C}$ except at isolated poles z_j inside $\mathcal{C}$, then

$$\int_{\mathcal{C}} dz\, f(z) = 2\pi i \sum_j \mathrm{Res}[f, z_j], \tag{2}$$

where the contour is traversed counterclockwise (if $\mathcal{C}$ is traversed clockwise, we get a minus sign on the right-hand side). For a simple pole at $z = z_0$, the residue is

$$\mathrm{Res}[f, z_0] = \lim_{z \to z_0} (z - z_0)\, f(z). \tag{3}$$

More general definitions include using the Laurent series for $f(z)$ around z_0.

2. **Jordan's lemma.** Let C_R be a semicircular arc of radius R in the upper half-plane. If $f(z) \to 0$ uniformly as $|z| \to \infty$ on C_R, then for any $a > 0$, the contour integral

$$\lim_{R\to\infty} \int_{C_R} dz\, f(z)\, e^{iaz} = \lim_{R\to\infty} \int_{C_R} dz\, f(z)\, e^{iaR\cos\theta} e^{-aR\sin\theta} \to 0, \tag{4}$$

where we wrote $z = Re^{i\theta}$ with $\theta \in [0, \pi]$ defining the semicircular arc. For $a < 0$, we draw the semicircle in the lower half-plane instead and take $\theta \in [\pi, 2\pi]$. In general, if an arbitrary integrand $f(z)$ decays sufficiently fast at infinity, the integral of $f(z)$ over a large semicircle vanishes.

3. **Laurent Series.** If $f(z)$ has an isolated singularity at z_0, it can be expanded in a Laurent series around z_0:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \tag{5}$$

where the coefficients a_n are given by

$$a_n = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f(z)}{(z - z_0)^{n+1}} dz, \tag{6}$$

and $\mathcal{C}$ is a small contour around z_0. The residue at z_0 is the coefficient a_{-1} of the $(z - z_0)^{-1}$ term in the Laurent expansion. Note, however, that the Green's function has simple poles, so the residue can be computed directly without needing the full Laurent expansion.

Evaluating real integrals by contour integration.

Method 1) Real integrals $I = \int_{-\infty}^{\infty} dx\, f(x)$ with poles in the real axis can be evaluated by extending $f(x)$ to the complex plane, $f(z)$. In particular, if $f(z) \to 0$ sufficiently fast as $|z| \to \infty$ in the upper (lower) half-plane, we can use Jordan's lemma to show that

$$I = \int_{-\infty}^{\infty} dx\, f(x) = \lim_{R\to\infty} \left(\int_{\mathcal{C}} dz\, f(z) - \int_{C_R} dz\, f(z) \right) = \int_{\mathcal{C}} dz\, f(z) = \pm 2\pi i \sum_j \mathrm{Res}[f, z_j], \tag{7}$$

where $\mathcal{C}$ is a closed contour that runs along the real axis and closes in the upper (lower) half-plane, and z_j are the poles of $f(z)$ enclosed by $\mathcal{C}$. Note that the poles on the real x-axis have been shifted slightly above or below the real axis with the appropriate $\pm i\epsilon$ prescription, which determines whether they are included in the contour or not. At the end of the calculation, we take $\epsilon \to 0^+$ to recover the physical result (though such limit often requires some care to be applied to the functional form of the result, see below).

Method 2) Alternatively, we can evaluate the integral I by using the Sokhotski–Plemelj formula, which states that for a function $f(z)$ with a simple pole at $z = z_0$ on the real axis,

$$\lim_{\epsilon\to 0^+} \int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - z_0 \pm i\epsilon} = \mathcal{P} \int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - z_0} \mp i\pi f(z_0), \tag{8}$$

where $\mathcal{P}$ denotes the Cauchy principal value, which is the limit of the integral taken by approaching x_0 from above and below:

$$\mathcal{P} \int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - x_0} = \lim_{\delta\to 0^+} \left(\int_{-\infty}^{x_0-\delta} dx\, \frac{f(x)}{x - x_0} + \int_{x_0+\delta}^{\infty} dx\, \frac{f(x)}{x - x_0} \right). \tag{9}$$

Let us investigate the most relevant example for us. A pole on the real axis, such as $1/(x - x_0)$, makes $\int dx\, f(x)$ ill-defined. Shifting $x_0 \to x_0 - i\epsilon$ moves the pole just below the real axis, and the integral becomes well-defined. Take the following real integral as an example:

$$\int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - x_0 + i\epsilon} = \mathcal{P} \int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - x_0} - i\pi f(x_0). \tag{10}$$

Say $f(x)$ is a smooth function that doesn't vary much around x_0, so we can approximate $f(x) \approx f(x_0)$ in the neighborhood of the pole. Then, the integral can be approximated as

$$\int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - x_0 + i\epsilon} \approx f(x_0) \int_{-\infty}^{\infty} dx\, \frac{1}{x - x_0 + i\epsilon} = -i\pi f(x_0). \tag{11}$$

This is a very common approximation in scattering theory, where the integrand has a pole at the energy eigenvalue E_n of the intermediate state, and we are interested in the contribution from the neighborhood of that pole.

We can do similar things by drawing our contour of integration with a large semicircle in one of the plane, and then instead of shifting the poles into or outside the contour, we can draw the contour to go around the poles, which gives us the same result as the Sokhotski-Plemelj formula above. To see that, we can write the integral as

$$\int_{-\infty}^{\infty} dx\, \frac{f(x)}{x - x_0} = \lim_{\delta\to 0^+} \left(\int_{-\infty}^{x_0-\delta} dx\, \frac{f(x)}{x - x_0} + \int_{x_0+\delta}^{\infty} dx\, \frac{f(x)}{x - x_0} \right) + \lim_{\delta\to 0^+} \int_{C_\delta} dz\, \frac{f(z)}{z - x_0}, \tag{12}$$

where C_δ is a small semicircle of radius δ around x_0. The first two integrals give us the Cauchy principal value, while the integral over C_δ gives us the contribution from the pole, which can be evaluated using the residue theorem to give us $\pm i\pi f(x_0)$, where the sign depends on whether we go around the pole in the upper or lower half-plane. To see that, write

$$\int_{C_\delta} dz\, \frac{f(z)}{z-x_0} = \int_0^\pi d\theta\, i\delta e^{i\theta} \frac{f(x_0+\delta e^{i\theta})}{\delta e^{i\theta}} = if(x_0)\int_0^\pi d\theta = i\pi f(x_0), \tag{13}$$

where we used the fact that $f(z)$ is analytic[1] at $z = x_0$ to write $f(x_0 + \delta e^{i\theta}) \simeq f(x_0)$ for small δ.

[1] A function is analytic at a point if it is differentiable in a neighborhood around that point.